

B.Tech/M.Tech (Integrated) DEGREE EXAMINATION, JULY 2025

Fifth Semester

21CSC302J - COMPUTER NETWORKS

(For the candidates admitted during the academic year 2021-2022 to 2024-2025)

Note:

- i. Part - A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40<sup>th</sup> minute.  
ii. Part - B and Part - C should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 × 1 = 20 Marks)

Marks BL CO

Answer all Questions

- |                                                                                                                                                                                                                                                                                                                                                  |   |   |   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| 1. Consider a small shop owner wants to set-up a network that connects 4 computers with a common printer for better data sharing and centralized printing. Which network is best suited for this set-up<br>(A) Wide Area Network (WAN)<br>(B) Metropolitan Area Network (MAN)<br>(C) Personal Area Network (PAN)<br>(D) Local Area Network (LAN) | 1 | 3 | 1 |
| 2. _____ refers to the total time taken for an entire message to reach its destination which includes transmission, propagation, processing and queuing delays<br>(A) Throughput<br>(B) Latency<br>(C) Bandwidth<br>(D) Data Rate                                                                                                                | 1 | 1 | 1 |
| 3. Which transmission medium is most suitable for a computer network operating in an environment with high electrical interference?<br>(A) Unshielded Twisted Pair (UTP)<br>(B) Coaxial Cable<br>(C) Fiber Optic Cable<br>(D) Shielded Twisted Pair (STP)                                                                                        | 1 | 3 | 1 |
| 4. Which technology is commonly used in circuit-switched networks for efficient bandwidth allocation?<br>(A) Packet Switching<br>(B) Message Switching<br>(C) Frequency-Division Multiplexing<br>(D) Time-Division Multiplexing (TDM)<br>(FDM)                                                                                                   | 1 | 2 | 1 |
| 5. A network has an IP address of 172.16.0.0 with a subnet mask of 255.255.255.240. How many usable host addresses are available in each subnet?<br>(A) 16<br>(B) 14<br>(C) 30<br>(D) 62                                                                                                                                                         | 1 | 3 | 2 |
| 6. How many bits are used for the Network ID and Host ID in a Class B IPv4 address?<br>(A) 16 bits for Network ID, 16 bits for Host ID<br>(B) 24 bits for Network ID, 8 bits for Host ID<br>(C) 8 bits for Network ID, 24 bits for Host ID<br>(D) 10 bits for Network ID, 22 bits for Host ID                                                    | 1 | 2 | 2 |
| 7. A network is assigned the 192.168.10.0/25 subnet. How many usable host addresses are available in each subnet?<br>(A) 62<br>(B) 14<br>(C) 30<br>(D) 126                                                                                                                                                                                       | 1 | 3 | 2 |
| 8. Which network device is used to regenerate and amplify signals to extend the transmission distance without altering the data?<br>(A) Switch<br>(B) Router<br>(C) Repeater<br>(D) Gateway                                                                                                                                                      | 1 | 1 | 2 |

9. What is the primary purpose of a default route in IP routing? 1 1 3  
 (A) To route packets to all available networks (B) To define a route when no specific route is found in the routing table  
 (C) To block all unknown traffic (D) To segment the packets and forward in different routes
10. Which routing protocol is known for its fast convergence and is Cisco proprietary? 1 2 3  
 (A) EIGRP (B) OSPF  
 (C) RIP (D) BGP
11. In Distance Vector Routing, routers exchange information with: 1 1 3  
 (A) All routers in the network (B) Only the destination routers  
 (C) Their directly connected neighbors (D) A central routing server
12. What is the loopback address in IPv6? 1 1 3  
 (A) 127.0.0.1 (B) ::1  
 (C) FE80::1 (D) 255.255.255.0
13. What is the primary purpose of using Hamming Codes in data communication? 1 1 4  
 (A) Detect and correct single-bit errors (B) Detect and correct multiple-bit errors  
 (C) Increase data transmission speed (D) Reduce the number of transmitted bits
14. The window size in the Sliding Window ARQ protocol determines the 1 2 4  
 (A) packet size (B) number of packets that can be received  
 (C) number of packets that can be transmitted before receiving an acknowledgment (D) number of errors that can be corrected
15. Which technique is used in wireless networks to avoid hidden terminal problems? 1 2 4  
 (A) ALOHA (B) RTS/CTS  
 (C) CSMA/CD (D) TDMA
16. In the HDLC protocol, which type of frame is primarily used for transmitting user data? 1 1 4  
 (A) Supervisory Frame (S) Frame (B) Unnumbered Frame (U) Frame  
 (C) Control Frame (D) Information (I) Frame
17. Which transport layer protocol is used for real-time applications like VoIP and online gaming due to its low overhead? 1 2 5  
 (A) TCP (B) UDP  
 (C) FTP (D) HTTP
18. Which port number is used by the Telnet protocol? 1 1 5  
 (A) 21 (B) 22  
 (C) 23 (D) 25
19. Which HTTP method is used to retrieve metadata about a resource without fetching its actual content? 1 1 5  
 (A) TRACE (B) HEAD  
 (C) CONNECT (D) OPTIONS
20. What is the purpose of a URL in the World Wide Web? 1 1 5  
 (A) To create web applications (B) To encrypt network traffic  
 (C) To identify and locate a resource on the internet (D) To store web pages locally

#### PART - B (5 × 8 = 40 Marks)

Answer **all** Questions

Marks BL CO

21. (a) A company is planning to set up a new network and is considering two Star or Mesh topologies. The company requires high reliability, scalability and cost-effectiveness while ensuring minimal downtime. Compare the structure of these topologies and discuss their advantages and disadvantages in relation to the company's needs. Analyze their suitability and recommend the most suited topology with a clear justification. 8 3 1  
 (OR)  
 (b) In a rural area with limited infrastructure, establishing internet connectivity poses significant challenges. Analyze the various types of unguided transmission media that can be used to deliver internet access in such an environment. Discuss their physical properties, performance characteristics and practical applications. Based on these factors determine the most suitable technology for this scenario.
22. (a) Convert the given IP addresses from binary notation to dotted decimal notation and identify its class and the default subnet mask 8 3 2  
 1. 01101111 10100101 00011000 11001100 [2 Marks]  
 2. 10001100 11000001 11111111 00000000 [2 Marks]  
 3. 11000000 10101000 00000001 00000001 [2 Marks]  
 4. 11110000 10111111 01101101 00110011 [2 Marks]  
 (OR)  
 (b) Consider an organization that has been allocated with the address 192.168.10.0 and it needs to create exactly 4 subnets using FLSM. Perform the following tasks  
 1. Determine the new subnet mask needed to achieve the required number of subnets. [2 Marks]  
 2. Calculate the total number of valid hosts per subnet. [2 Marks]  
 3. Construct a table listing the Network IDs, First Host ID, Last Host ID, Broadcast ID [4 Marks]
23. (a) Explain the route method and next-hop forwarding in detail, highlighting their differences with a suitable diagram. 8 2 3  
 (OR)  
 (b) Outline the concept of Distance Vector Routing, explaining its working principles with an example and appropriate diagrams.
24. (a) Determine the number of check bits required for a 16-bit message when using Hamming code for single-bit error detection and correction. Additionally, construct the transmitted bit pattern for the message 1101001100110101, assuming even parity is used in the Hamming code. 8 3 4  
 (OR)  
 (b) With the message M=1100101110, find the CRC remainder using the divisor polynomial  $x^4+x^2+x+1$ . Describe how you would verify if an error occurred during transmission.
25. (a) Illustrate the transmission mechanism of UDP and TCP protocols. 8 2 5  
 (OR)  
 (b) Describe the functionalities of HTTP and FTP, focusing on their working principles and key features.

#### PART - C (1 × 15 = 15 Marks)

Answer **any 1** Questions

Marks BL CO

26. In a network communication scenario, a sender transmits data to a receiver across multiple layers. Using the ISO-OSI network architecture, explain how the data flows through each layer, highlighting the key functionalities at every stage. 15 2 1
27. Explain the ARQ protocol used for error control and how it ensures reliable data transmission. Also, list and describe the different types of ARQ mechanisms. 15 2 4

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